

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12403549
B2
September 02, 2025
Shapiro; Daniel et al.

Mechanical system for high positional computer numerically controlled applications

Abstract

Disclosed embodiments include a gantry assembly that has (i) a moveable carriage with a laser head affixed thereto and (ii) two sides (moveable subassemblies) that are connected by a joining subassembly. One side of the gantry has two motors: (i) a first motor (x-axis motor) that moves the carriage along a first axis (x-axis) between the two sides of the gantry, and (ii) a second motor (y-axis motor) that moves the gantry along a second axis (y-axis) perpendicular to the first axis. In some embodiments, the gantry assembly also includes a drive shaft (or alternative drive mechanism) operated by the second motor (y-axis motor) to facilitate movement of the two sides of the gantry together along the second axis (y-axis).

Inventors: Shapiro; Daniel (Seattle, WA), Gosselin; Mark (Seattle, WA), Webster; Douglas (Seattle, WA)

Applicant: Glowforge, Inc. (Seattle, WA)

Family ID: 86055418

Assignee: Glowforge, Inc. (Seattle, WA)

Appl. No.: 17/511000

Filed: October 26, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230128807 A1	Apr. 27, 2023

Publication Classification

Int. Cl.: B23K26/08 (20140101)

U.S. Cl.:

CPC **B23K26/0876** (20130101); G05B2219/33198 (20130101); G05B2219/40293 (20130101)

Field of Classification Search

CPC: B23K (26/0876); B23K (26/0869); B23K (26/10); G05B (2219/33198); G05B (2219/40293)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3736402	12/1972	Mefferd	901/41	B23K 26/042
4985780	12/1990	Garnier	219/121.68	B23K 26/10
5051558	12/1990	Sukhman	219/121.75	B23K 26/128
9266193	12/2015	Liu	N/A	B23K 26/0665
2003/0034336	12/2002	Erlenmaier	219/121.67	B23K 26/0876
2007/0034613	12/2006	Lundberg	219/121.68	B23K 26/0876
2019/0111525	12/2018	Okamura	N/A	H02K 41/031
2021/0121985	12/2020	Rubens	N/A	B23K 37/0538
2021/0237205	12/2020	Liu	N/A	B23K 26/38

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
208276343	12/2017	CN	N/A
109146781	12/2018	CN	N/A
210651907	12/2019	CN	N/A
109048072	12/2019	CN	N/A
111515523	12/2019	CN	N/A
113971661	12/2021	CN	N/A
215846437	12/2021	CN	N/A
108846819	12/2021	CN	N/A

Primary Examiner: Elahmadi; Zakaria

Attorney, Agent or Firm: Akerman LLP

Background/Summary

FIELD OF THE DISCLOSURE

(1) The subject matter described herein relates to controlling the position of a laser in a computer numerically controlled (CNC) machine configured for laser cutting, laser drilling, laser ablation, laser engraving, and laser machining applications. Aspects of the disclosed subject matter may additionally relate to other types of CNC machines as well.

BACKGROUND

(2) A computer numerically controlled (CNC) machine operates by moving a tool over a material to be machined. In some CNC machines, moving the tool over the material to be machined comprises

moving a laser or similar energy source configured to deliver electromagnetic energy to one or more locations along the material to be machined.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Features, aspects, and advantages of the presently disclosed technology may be better understood with regard to the following description, appended claims, and accompanying drawings, as listed below. A person skilled in the relevant art will understand that the features shown in the drawings are for purposes of illustrations, and variations, including different and/or additional features and arrangements thereof, are possible.
- (2) FIG. 1 shows aspects of a computer numerically controlled (CNC) machine according to some embodiments.
- (3) FIG. 2A shows a top view of a gantry assembly.
- (4) FIG. 2B shows a top view of a gantry assembly.
- (5) FIG. 3A shows a top view of a gantry assembly according to some embodiments.
- (6) FIG. 3B shows a top view of aspects of a gantry assembly according to some embodiments.
- (7) FIG. 4A shows aspects of a carriage used with gantry assemblies according to some embodiments.
- (8) FIG. 4B shows a laser head enclosure affixed to a carriage and configured to house the laser head according to some embodiments.
- (9) FIG. 4C shows an angle adapter for use with the carriage according to some embodiments.
- (10) FIG. 4D shows a laser head enclosure attached to an angle adapter according to some embodiments.
- (11) The drawings are for the purpose of illustrating example embodiments, but those of ordinary skill in the art will understand that the technology disclosed herein is not limited to the arrangements and/or instrumentality shown in the drawings.

DETAILED DESCRIPTION

I. Overview

- (12) Some types of computer numerically controlled (CNC) machines such as CNC machines for laser cutting, laser drilling, laser ablation, laser engraving, laser machining, laser-based three-dimensional (3D) printers, and other types of CNC machines operate by moving a laser (or similar electromagnetic emission source) over a material to be machined, thereby burning, cutting, ablating, vaporizing, melting, or otherwise removing portions of the material to be machined. In connection with moving the laser over the material to be machined, it is desirable to have precise control over the movement, alignment, and positioning of the laser over the material to be machined because a more precisely controlled laser enables the CNC machine to produce a more precisely machined material and thus, a higher quality end product as compared to CNC machines with less precisely controlled lasers.
- (13) Typical CNC machines control the movement, alignment, and positioning of the laser over the material to be machined via an arrangement of components that can add cost and complexity to the CNC machine. For example, typical CNC machines include multiple stepper motors and electronic systems configured to drive a gantry assembly arranged to control the movement, alignment, and positioning of the laser over the material to be machined with the CNC machine. These sets of motors, controllers, and gantry structures often also require fixtures that can limit the functionality of the CNC machine in some respects. For example, typical gantry assemblies require stabilizing beams, shafts, and/or other support structures that can limit the range of movement of the gantry assembly, and thus, the laser head. These stabilizing beams, shafts, and/or other support structures can also obstruct material pass-through areas or other regions within the operating area of the CNC

machine, thereby limiting the size of the materials that the CNC machine can process (i.e., limiting the CNC machine's ability to accept larger pieces of material for processing within the CNC machine).

(14) Embodiments disclosed and described herein overcome or at least ameliorate some of the above-described shortcomings of typical CNC machines by implementing a new gantry assembly that provides precise control over the movement, alignment, and positioning of the laser over the material to be machined, but with fewer components and stabilizing beams, shafts, and/or other support structures as compared to CNC machines with traditional gantry assemblies.

(15) For example, some embodiments include a gantry that has (i) a moveable carriage with a laser head affixed thereto and (ii) two sides (moveable subassemblies) that are connected by a joining subassembly. One side of the gantry has two motors: (i) a first motor (x-axis motor) that moves the carriage back and forth across the joining subassembly along a first axis (x-axis) between the two sides of the gantry, and (ii) a second motor (y-axis motor) that moves the gantry along a second axis (y-axis) substantially perpendicular to the first axis. In some embodiments, the gantry also includes a drive shaft (or alternative drive mechanism) operated by the second motor (y-axis motor) to facilitate movement of the two sides of the gantry together along the second axis (y-axis). In embodiments where the two motors are mounted on the same side of the gantry, the position of the first motor relative to the second motor remains constant in operation while both (i) the carriage moves back and forth across the joining subassembly along the x-axis and (ii) the two sides of gantry move together along the y-axis, thereby moving the carriage along the y-axis.

(16) While some examples described herein may refer to functions performed by given actors such as “users” and/or other entities, it should be understood that this is for purposes of explanation only. The claims should not be interpreted to require action by any such example actor unless explicitly required by the language of the claims themselves.

(17) In the Figures, identical reference numbers typically identify generally similar, and/or identical, elements. Many of the details, dimensions, angles and other features shown in the Figures are merely illustrative of particular embodiments of the disclosed technology. Accordingly, other embodiments can have other details, dimensions, angles and features without departing from the spirit or scope of the disclosure. In addition, those of ordinary skill in the art will appreciate that further embodiments of the various disclosed technologies can be practiced without several of the details described below.

II. Example CNC Machine Embodiments

(18) FIG. 1 shows aspects of a computer numerically controlled (CNC) machine **100** according to some embodiments. The CNC machine **100** includes a housing **102**, an interior space **104**, and a lid **106**. The interior space **104** of the housing **102** is defined by four interior walls **108**, **110**, **112**, and **114**. Interior wall **110** is along the left side of the interior space **104**, interior wall **108** is along the right side of the interior space **104**, interior wall **114** is along the back side of the interior space **104** below where the lid **106** joins with the housing **102**, and interior wall **112** is along the front side of the interior space **104** on the inside of a front cover plate **118** where the lid **106** closes to enclose the interior space **104**. CNC machine **100** also includes pass-through slots **120** and **121** that enables materials that are to be machined to be slid through the front cover plate **118** and back wall **114**. In some embodiments, the front cover plate **118** may be removable to create a larger area, which facilitates machining of materials that may be too large to fit through the pass-through slot **120**. In alternate embodiments, pass-through slots may be positioned on the side walls **110** and **112** instead of or in addition to the pass-through slots **120** and **121** that are positioned on the front and rear walls **108** and **114**, respectively. However, the interior space in other embodiments may be defined by more or fewer interior walls configured differently than shown in the example illustrated in FIG. 1.

(19) The interior space **104** is arranged to accommodate a gantry assembly (e.g., gantry assembly **300** shown in FIG. 3A) within the interior space **104**. The gantry assembly is arranged to control

the movement, alignment, and positioning of a laser head arranged to apply laser energy to materials placed within the interior space **104** of the housing **102** of the CNC machine **100**. In operation, material to be machined (not shown) is placed into the interior space **104** of the housing **102**, the lid **106** is closed, and the CNC machine **100** controls (via the gantry assembly, e.g., gantry assembly **300**) the movement, alignment, and positioning of the laser head over the material to be machined within the interior space **104** of the housing **102**.

(20) CNC machine **100** also includes a control compartment **116** that is configured to house electrical components for powering and/or controlling the CNC machine **100**. By housing the electrical components within the control compartment **116**, the electrical components are protected from the conditions (e.g., heat, smoke, laser energy reflections) inside the interior space **104** where the laser operates.

(21) For example, in some embodiments, control compartment **116** is configured to house one or more processors and tangible, non-transitory computer readable media (e.g., computer memory). In some configurations, the one or more processors and/or the tangible, non-transitory computer readable media are mounted on one or more circuit boards within the control compartment **116**. In some embodiments, one or more processors are configured to communicate with sensors located in the interior space **104**, on the gantry assembly, and/or elsewhere on or within the housing **102** to collect information on the operation of the CNC machine **100** that can be used for controlling various aspects of the CNC machine **100**, such as operating fans or operating safety features of the CNC machine **100** (e.g., safety shutdown).

(22) In some embodiments, the control compartment **116** additionally includes one or more communications interfaces configured to communicate with one or more external computing systems (e.g., a personal computer, laptop, smartphone, tablet computer, and/or Internet-based server system). The one or more communications interfaces may be any type of wireless and/or wired communications interface, e.g., WiFi, Bluetooth, Ethernet, or any other type of communications interface now known or later developed that is suitable for transmitting and receiving data between the CNC machine **100** and the one or more external computing systems. In operation, the one or more computing systems include their own one or more processors and tangible, non-transitory computer readable media with instructions for communicating with the CNC machine **100** and controlling the operation of the CNC machine **100**.

(23) In operation, the one or more processors of the CNC machine **100** execute program instructions stored on the tangible, non-transitory computer readable media of the CNC machine **100**. In some embodiments, the program instructions include instructions for controlling movement of the gantry (and thus, the laser head) within the interior space **104** of the housing **102** via motors, gears, and pulleys of the gantry assembly as described in further detail herein. In some embodiments, the control compartment **116** may additionally house power supplies and related electronics for powering and/or controlling the electric motors that control the movement and positioning of the gantry (and thus, the carriage **302** and ultimately the laser head).

III. Typical Gantry Assemblies

(24) FIG. 2A shows a top view of a gantry assembly **200**. In operation, gantry assembly **200** may be installed within a CNC machine. For example, gantry assembly **200** may be installed within the interior space **104** of the housing **102** of the CNC machine **100**.

(25) Gantry assembly **200** is a linear system that includes two y-rails **202** and **204** to support the motion of a gantry **206** in the y-direction. The gantry **206** (e.g., formed from extruded metal) spans the distance between the two y-rails **202** and **204** to support the motion of the laser head **208** (affixed to the gantry **206**) in the x-direction.

(26) Gantry assembly **200** also includes two separate y-motor/belt/idler pulley subassemblies, where each y-motor/belt/idler pulley subassembly is attached to one of the y-rails **202** and **204** and arranged to control the motion of the gantry **206** in the y-direction. In particular, y-motor **210** (e.g., a stepper motor) is coupled to y-gear **214**, which drives belt **218**, which runs between the y-gear

214 and idler pulley **222**. The y-motor/belt/idler pulley subassembly formed from y-motor **210**, y-gear **214**, belt **218**, and idler pulley **222** is attached to y-rail **202**. Similarly, y-motor **212** (e.g., a stepper motor) is coupled to y-gear **216**, which drives belt **220**, which runs between the y-gear **216** and idler pulley **224**. The y-motor/belt/idler pulley subassembly formed from y-motor **212**, y-gear **216**, belt **220**, and idler pulley **224** is attached to y-rail **204**. In operation, the y-motor **212** and the y-motor **210** must operate in a closely coordinated manner to accurately control the movement and positioning of the gantry **206**. The arrangement of gearing and mechanical components to closely coordinate the operation of y-motor **212** and y-motor **210** adds cost and complexity to the CNC machine.

(27) Gantry assembly **200** also includes x-motor **226** (e.g., a stepper motor), coupled to an x-gear **228**, which drives belt **230**, which runs between x-gear **228** and idler pulley **232**. In operation, the x-motor **226**, x-gear **228**, belt **230**, and idler pulley **232** work in concert to control the motion of the laser head **208** in the x-direction by moving the laser head **208** (which includes the laser (not shown)) back and forth along the gantry **206** between y-rail **202** and y-rail **204**.

(28) Each of the y-motor **210**, y-motor **212**, and x-motor **226** requires its own power and control signals and corresponding wiring routed thereto (not shown).

(29) One typical design goal of a laser system is a lightweight gantry. A lightweight gantry can be moved at satisfactory speeds for laser applications with lower powered electric motors as compared to a heavier gantry, which would require higher powered (and therefore more expensive) motors to move the heavier gantry at the same speed as the lightweight gantry. Because electric motors tend to be one of the heavier individual components of a gantry drive system, the gantry assembly **200** attempts to satisfy the design goal of a lightweight gantry by locating only the x-motor **226** on the gantry **206** and locating the y-motors **210** and **212** off of the gantry on y-rails **202** and **204**.

(30) FIG. 2B shows a top view of another gantry assembly **201**. In operation, gantry assembly **201** may be installed within a CNC machine. For example, gantry assembly **201** may be installed within the interior space **104** of the housing **102** of the CNC machine **100**.

(31) Gantry assembly **201** (FIG. 2B) is the same as gantry assembly **200** (FIG. 2A) in all respects except that (i) gantry assembly **201** has only a single y-motor (i.e., y-motor **234**) rather than two y-motors (i.e., y-motors **210** and **212**) as in gantry assembly **200**, and (ii) gantry assembly **201** includes shaft **236** connecting y-gear **216** with y-gear **214** whereas gantry assembly **200** does not contain shaft **236**.

(32) In operation, the single y-motor **234** is connected to a set of y-gear/belt/idler pulley subassemblies via shaft **236**. Each y-gear/belt/idler pulley subassembly is attached to one of the y-rails **202** and **204** and arranged to control the motion of the gantry **206** in the y-direction. In particular, y-motor **234** (e.g., a stepper motor) is coupled via shaft **236** to (i) y-gear **216**, which drives belt **220**, which runs between y-gear **216** and idler pulley **224**, and (ii) y-gear **214**, which drives belt **218**, which runs between y-gear **214** and idler pulley **222**. The y-gear/belt/idler pulley subassembly formed from y-gear **214**, belt **218**, and idler pulley **222** is attached to y-rail **202**. The y-gear/belt/idler pulley subassembly formed from y-gear **216**, belt **220**, and idler pulley **224** is attached to y-rail **204**.

(33) Another typical design goal of some laser systems is a low-cost gantry drive system. Because one of the more expensive individual components of a gantry drive system is the electric motors, the gantry assembly **201** attempts to satisfy the design goal of a low cost gantry drive system by controlling the gantry **206** with only two motors (i.e., x-motor **226** and y-motor **234**) as compared to the gantry assembly **200** (FIG. 2A) which uses three motors (i.e., x-motor **226**, y-motor **210**, and y-motor **212**).

(34) However, using a single y-motor **234** as in gantry assembly **201** can negatively impact performance of the CNC machine **100**. For example, movement of the gantry **206** along y-rail **202** (the side of the gantry assembly **201** without a y-motor) tends to trail movement of the gantry **206** along y-rail **204** (the side of the gantry assembly **201** where y-motor **234** is positioned) primarily

because of the weight of the gantry **206** and the distance between the y-rails **202** and **204**. Further, the speed of the gantry **206** in the y-direction produces torque that exacerbates the problem because the shaft **236** has mass that tends to produce a trailing effect on the positioning of the gantry **206**, and thus the laser head **208** comprising the laser (not shown). This torque can both (i) limit the speed at which the gantry assembly **201** can move the gantry **206** (and thus the laser head **208** comprising the laser) at least along the y-direction, and/or (ii) reduce the alignment precision of the gantry **206** (and thus the laser head **208** comprising the laser). In some scenarios, the potential for misalignment error increases at laser head positions located further away from y-motor **234**.

(35) In addition to reduced speed and lower precision as compared to gantry assembly **200** (FIG. 2A), the addition of shaft **236** in gantry assembly **201** (FIG. 2B) to enable the y-motor **234** to additionally drive the y-gear **214** also introduces an obstruction (i.e., the shaft **236** itself) that can prevent the placement of larger material into the interior space **104** of the housing **102** of a CNC machine **100** (FIG. 1). For example, the shaft **236** may obstruct the insertion of materials through pass-through slots **120** and **121** (FIG. 1). Or in embodiments where the front plate **118** is removable, the shaft **236** may obstruct or otherwise restrict the ability to place larger material into the interior space **104** (FIG. 1).

IV. Example Gantry Assembly Embodiments

(36) FIG. 3A shows a top view of a gantry assembly **300** according to some embodiments of the inventions disclosed and described herein. In operation, gantry assembly **300** may be installed within a CNC machine such as CNC machine **100** (FIG. 1). For example, and as described in more detail below, gantry assembly **300** may be installed within the interior space **104** of the housing **102** of the CNC machine **100**.

(37) Gantry assembly **300** includes a carriage **302** that houses a laser head **304**, a right side subassembly **306**, and a left side subassembly **308**. The right side subassembly **306** is moveable in the y-direction along y-rail **326** and the left side subassembly **308** is movable in the y-direction along y-rail **328**. The right side subassembly **306** is joined to the left side subassembly **308** via the joining subassembly **310** such that the right side subassembly **306**, the left side subassembly **308**, and the joining subassembly **310** move together as a single gantry structure in manner similar to gantry **206** (FIGS. 2A and 2B).

(38) When the gantry assembly **300** is installed within the CNC machine **100** of FIG. 1, the y-rail **326** (on the right side of the gantry assembly **300**) is proximate to and extends substantially along at least a portion of the right interior wall **108** of the housing **102**, and the y-rail **328** (on the left side of the gantry assembly **300**) is proximate to and extends substantially along at least a portion of the left interior wall **110** of the housing **102** opposite the right interior wall **108**.

(39) In some embodiments, the y-rail **326** extends substantially along the right interior wall **108** of the housing **102** along the length (or substantially along the length) from interior wall **112** at the front of the housing **102** to interior wall **114** at the back of the housing **102**. And the y-rail **328** extends substantially along the left interior wall **110** of the housing **102** along the length (or substantially along the length) from interior wall **112** at the front of the housing **102** to interior wall **114** at the back of the housing **102**. In some embodiments, the interior walls of the housing **102** include notches or other indentations along the walls that are designed to receive the y-rails **326** and **328** (or at least mounting portions thereof), thereby enabling the y-rails **326** and **328** to be snapped and/or plugged securely into place via the notches or other indentations.

(40) In operation, the right side subassembly **306** is configured to move in the y-direction within the interior space **104** of the housing **102** via y-rail **326**, and the left side subassembly **308** is configured to move in the y-direction within the interior space **104** of the housing **102** via y-rail **328**. The y-rail **328** (on the left side) is substantially parallel to y-rail **326** (on the right side). As mentioned earlier, the right side subassembly **306** is joined to the left side subassembly **308** via the joining subassembly **310** such that the right side subassembly **306**, the left side subassembly **308**, and the joining subassembly **310** move together as a single gantry structure.

(41) In some embodiments, y-rail **326** and y-rail **328** are made from a hard, low friction material. In operation, y-rail **326** and y-rail **328** can be used to align the gantry assembly **300** and restrict degrees of freedom of movement of the gantry assembly **300** to help maintain positional accuracy of gantry assembly **300** components, including but not limited to the carriage **302** and the laser head **304** affixed thereon.

(42) In some embodiments, the right side subassembly **306** includes one or more friction pads (or similar friction causing components now known or later developed) at the interface between the bottom of the right side subassembly **306** and the top of the y-rail **326**. Additionally or alternatively, the left side subassembly **308** includes one or more friction pads (or similar friction causing components now known or later developed) at the interface between the bottom of the left side subassembly **308** and the top of the y-rail **328**. These friction pads (or similar components) maintain a constant (or substantially constant) preload force to prevent misalignment of the gantry structure (formed from the right side subassembly **306**, left side subassembly **308**, and the joining subassembly **310**) while in motion, particularly when the gantry structure starts to move in the y-direction along y-rails **326** and **328** and when the gantry structure slows to a stop in the y-direction along y-rails **326** and **328**. In this manner, the one or more friction pads (or similar components) form at least one of (i) a first interface between the right side subassembly **306** and y-rail **326** that provides a first preload force between the right side subassembly **306** and y-rail **326**, and (ii) a second interface between the left side subassembly **308** and y-rail **328** that provides a second preload force between the left side subassembly **308** and the y-rail **328**.

(43) Additionally, some embodiments include anti-lift and/or stabilization mechanisms located under one or both of the y-rail **326** and/or y-rail **328** that prevent the gantry structure from lifting (e.g., via user interference or other force) away from the y-rail **326** and/or y-rail **328**. These anti-lift and/or stabilization mechanisms may take the form of one or more clips, locks, latches, hooks, or other mechanisms sufficient to prevent the gantry structure of the gantry assembly **300** from moving in the z-direction, perpendicular to the x-direction and the y-direction. Lifting or otherwise moving one or both of the right side subassembly **306** or the left side subassembly **308** away from the y-rail **326** or y-rail **328** can cause the gantry assembly **300** components to become misaligned and/or cause the gantry assembly **300** control algorithms to become uncalibrated, either of which can cause poor laser fabrication results.

(44) In some embodiments, a first anti-lift and/or stabilization mechanism is located on the right side subassembly **306** (and moves with the right side subassembly **306**) and a second anti-lift and/or stabilization mechanism is located on the left side subassembly **308** (and moves with the left side subassembly **308**). In operation, these one or more anti-lift and/or stabilization mechanisms are configured to constrain movement of the right side subassembly **306** and the left side subassembly **308** in the z-direction, substantially perpendicular to both the x-direction and the y-direction.

(45) As mentioned earlier, the right side subassembly **306** and the left side subassembly **308** are joined via a joining assembly **310** that extends between the right side subassembly **306** and the left side subassembly **308**. The fixed shaft **318** rigidly connects the right side subassembly **306** to the left side subassembly **308**. In some embodiments, the fixed shaft **318** is substantially parallel to the drive shaft **316**.

(46) In the example embodiments shown in FIGS. **3A** and **3B**, the joining assembly **310** includes a fixed shaft **318** and a drive shaft **316**. However, in other embodiments, the joining assembly **310** may include joining mechanisms different than cylindrically-shaped shafts. For example, rather than a cylindrically-shaped shaft with a circular cross-section, one or both of the fixed shaft **318** and/or the drive shaft **316** may have an oval-shaped cross-section, a square-shaped cross-section, a rectangular-shaped cross section, etc.

(47) Ultimately, the fixed shaft **318** may take substantially any form or shape that is sufficient to maintain a rigid physical connection between the right side subassembly **306** and the left side subassembly **308** that causes the right side subassembly **306**, the left side subassembly **308**, and the

joining subassembly **310** to move together as a single component or structure (sometimes referred to herein as the gantry structure of gantry assembly **300**). Additional details regarding the configuration and arrangement of the fixed shaft **318**, drive shaft **316**, carriage **302**, and laser head **304** in various embodiments are shown and described herein with reference to FIGS. **4A-4D**.

(48) In some alternative embodiments, the joining subassembly **310** that joins the left side subassembly **308** with the right side subassembly **306** may additionally or alternatively include (i) a fixed member that connects the right and left side subassemblies in a manner similar to the fixed shaft **318**, and (ii) a hollow member arranged to at least partially enclose a drive shaft or similar drive mechanism. In such alternative embodiments, the carriage **302** is configured to slide along the fixed member and the hollow member back and forth along the joining subassembly **310** in the x-direction in a manner similar to how the carriage **302** is described as sliding along the fixed shaft **318** and the drive shaft **316**.

(49) In FIG. **3A**, the left side subassembly **308** includes (i) an x-motor **312** that is configured to control the position of the carriage **302** (and the laser head **304** affixed thereto) in the x-direction along the joining subassembly **310** between the right side subassembly **306** and the left side subassembly **308**, and (ii) a y-motor **314** that is configured to control the position of the right side subassembly **306** and the left side subassembly **308** together along y-direction. As shown in FIG. **3A**, the x-direction is substantially perpendicular to the y-direction. In some embodiments, the x-motor **312** and y-motor **314** are positioned at substantially 90 degrees relative to each other on the left side subassembly **308**. In some embodiments, the x-gear **324** and y-gear **322** (described further below) are additionally or alternatively positioned at substantially 90 degrees relative to each other on the left side subassembly **308**.

(50) The drive shaft **316** is connected to and operated by the y-motor **314**. The drive shaft **316** extends along at least a portion of the joining subassembly **310** and is arranged to facilitate movement of the right side subassembly **306** together with the left side subassembly **308** in the y-direction.

(51) To facilitate the movement of the right side subassembly **306** together with the left side subassembly **308** in the y-direction, the right side subassembly **306** includes y-gear **320** connected to the drive shaft **316**, and the left side subassembly **308** includes a corresponding y-gear **322**. The y-gear **320** on the right side subassembly **306** and the y-gear **322** on the left side subassembly **308** are connected to each other via the drive shaft **316**, which is coupled to the y-motor **314** via a linkage (not shown). In operation, the y-motor **314** is configured to (i) control movement of the y-gear **322** (on the left side subassembly **308**) via the linkage (not shown) and (ii) control movement of the y-gear **320** (on the right side subassembly **306**) by axially rotating the y-gear **322**, which in turn axially rotates the drive shaft **316** and the y-gear **320** (on the right side subassembly **306**) connected thereto.

(52) Also to facilitate the movement of the right side subassembly **306** together with the left side subassembly **308** in the y-direction, the gantry assembly **300** additionally includes two belt-and-pulley arrangements that are driven by y-motor **314**, y-gear **322**, drive shaft **316**, and y-gear **320**. In the first belt-and-pulley arrangement (on the right side), y-gear **320** is connected to and drives belt **330**, which runs between idler pulley **332** and idler pulley **334**. This first belt-and-pulley arrangement is configured to facilitate movement of the right side subassembly **306** in the y-direction via y-rail **326**. In the second belt-and-pulley arrangement (on the left side), y-gear **322** is connected to and drives belt **338**, which runs between idler pulley **336** and idler pulley **340**. This second belt-and-pulley arrangement is configured to facilitate movement of the left side subassembly **308** in the y-direction via y-rail **328**.

(53) In alternative embodiments, rather than the first and second belt-and-pulley arrangements, y-motor **314** and y-gears **320** and **322** may instead facilitate movement of the right side subassembly **306** and left side subassembly **308** via an alternative mechanism, such as a screw drive, chain drive, rack gearing or other type of drive and/or gearing mechanism now known or later developed that is

sufficient to facilitate movement of the right side subassembly **306** and left side subassembly **308** along the y-rails **326** and **328**.

(54) The left side subassembly **308** also includes x-gear **324**, which is coupled to the x-motor **312** via a second linkage (not shown). The x-motor **312** is configured to control the position of the carriage **302** in the x-direction along the joining subassembly **310** via the second linkage (not shown) and the x-gear **324**. The x-motor **312** and x-gear **324** work in conjunction with a third belt-and-pulley arrangement including belt **342** and idler pulley **344**. In this third belt-and-pulley arrangement, the x-gear drives belt **342**, which runs between x-gear **324** and idler pulley **344**. This third belt-and-pulley arrangement is configured to facilitate movement of the carriage **302** in the x-direction along the joining subassembly **310** extending between the right side subassembly **306** and left side subassembly **308**.

(55) In alternative embodiments, rather than the third belt-and-pulley arrangement, x-motor **312** and x-gear **324** may instead control the position of the carriage **302** in the x-direction along the joining subassembly **310** via an alternative mechanism, such as a screw drive, chain drive, or other type of drive mechanism now known or later developed that is sufficient to move the carriage **302** back and forth along the joining subassembly **310** in the x-direction between the right side subassembly **306** and the left side subassembly **308**.

(56) FIG. **3B** shows a top view of aspects of a gantry assembly **301** according to some alternative embodiments. Like the gantry assembly **300** shown in FIG. **3A**, the gantry assembly **301** shown in FIG. **3B** may be installed within a CNC machine such as CNC machine **100** (FIG. **1**) in substantially the same manner as gantry assembly **300** shown in FIG. **3A**.

(57) Gantry assembly **301** is substantially the same as gantry assembly **300** in all respects and includes many of the same components as gantry assembly **300** except that gantry assembly **301** in FIG. **3B** uses a right rack gear assembly **360** integrated with the right y-rail **326** and a left rack gear assembly **362** integrated with the left y-rail **328** to facilitate the movement of the right side subassembly **306** together with the left side subassembly **308** in the y-direction rather than the two belt-and-pulley arrangements used in gantry assembly **300** shown in FIG. **3A**. In operation, the right gear rack assembly **360** and left gear rack assembly **362**, in conjunction with the drive shaft **316**, helps to maintain the squareness of the gantry assembly **301**, while also requiring fewer components (and thus, having a corresponding lower cost) than the gantry assembly **300** shown in FIG. **3A**.

(58) In some embodiments, the x-motor **312** is a stepper motor and the y-motor **314** is a stepper motor. In other embodiments, the x-motor **312** and/or the y-motor **314** may be any other type of electric motor now known or later developed that is suitable for controlling the x-y positioning of the carriage **302** for CNC applications.

(59) In some scenarios, dynamic problems can arise where rotation of the drive shaft **316** causes one side or corner of the right side subassembly **306** to lift up, which can affect positioning accuracy of the gantry assembly **300/301**. Therefore, in some embodiments, the right side subassembly **306** is weighted to counteract the lifting force imposed upon the right side subassembly **306** caused by rotation of the drive shaft **316**. For example, if the weight of the right side subassembly **306** and all the components located thereon (or therein) is not sufficient to counteract the lifting force caused by rotation of the drive shaft **316**, then some embodiments may include one or more weights placed strategically on (or perhaps within) the right side subassembly **306** to counteract the lifting force and, in turn, improve positioning accuracy.

(60) In operation, the carriage **302** (and the laser head **304** affixed thereto) moves in the x-direction along the length of the joining assembly **310**, and the joining assembly **310** moves in the y-direction together with the right side subassembly **306** and the left side subassembly **308**, thereby moving the carriage **302** (and the laser head **304** affixed thereto) in the y-direction. This combination of x-direction movement and y-direction movement enables the carriage **302** (and the laser head **304**) to be positioned at different x-y coordinates within the interior space **104** of the

housing **102** of the CNC machine **100** (FIG. 1). In some embodiments, the laser head **304** includes components, e.g., motors and lenses (not shown), that are configured to focus the laser contained therein at different positions along the z-direction within the interior space **104** of the housing **102** of the CNC machine **100**.

(61) FIG. 4A shows aspects of the carriage **302** used with gantry assemblies **300/301** according to some embodiments. The carriage **302** is configured to support a laser head contained within a laser head enclosure **406**. FIG. 4B shows the laser head enclosure **406** affixed to the carriage **302** and configured to house the laser head (e.g., laser head **304** shown in FIGS. 3A and 3B).

(62) As explained above with reference to FIGS. 3A and 3B, the x-motor **312** and x-gear **324** work in conjunction with a belt-and-pulley arrangement including belt **342** and idler pulley **344**. In this belt-and-pulley arrangement, the x-gear drives belt **342**, which runs between x-gear **324** and idler pulley **344**. This belt-and-pulley arrangement is configured to facilitate movement of the carriage **302** in the x-direction along the fixed shaft **318** and drive shaft **316** of the joining subassembly **310** extending between the right side subassembly **306** and left side subassembly **308**.

(63) In the embodiment shown in FIG. 4A, belt **342** runs parallel to the fixed shaft **318** and the drive shaft **316**. The belt **342** is routed with respect to the fixed shaft **318** and drive shaft **316** so that an opening exists between the fixed shaft **318** and drive shaft **316**. Once the laser head enclosure **406** is attached to the carriage **302** as shown in FIG. 4B, an opening in the bottom of the carriage **302** aligns with the opening between the fixed shaft **318** and drive shaft **316**. This alignment allows the laser contained within the laser head enclosure **406** to fire downward through the opening in the bottom of the carriage **302** and between the fixed shaft **318** and drive shaft **316** toward material positioned within the interior space **104** of the housing **102** (FIG. 1) of the CNC machine **100** (FIG. 1) below the gantry assembly **300/301**.

(64) In some embodiments, the laser head enclosure **406** includes a flat flex cable opening **404** arranged to receive a flat flex cable configured to provide components contained within the laser head enclosure **406** (e.g., the laser head and/or other components) with power and control signaling (and perhaps other data, too) from corresponding power and control circuitry, respectively. In some embodiments, at least some of the power and control circuitry for powering, communicating with, and controlling the components contained within the laser head enclosure **406** are housed in the control compartment **116** of the CNC machine **100** (FIG. 1). Although a flat flex cable (sometimes called a ribbon cable) is shown in the example embodiment depicted in FIG. 4B, any type of cabling now known or later developed could be used to provide power, communication, and control signaling to components within the laser head enclosure **406**.

(65) Alternatively, in some embodiments, rather than using a physical cable to provide power, control signaling, and/or other data to/from the components contained within the laser head enclosure **406**, power, control signaling, and/or other data could instead be provided to (or received from) the components within the laser head enclosure **406** via one or both of the drive shaft **316** and/or the fixed shaft **318**.

(66) For example, in some embodiments, power and/or control signaling components housed in control compartment **116** are connected to power and/or control interfaces located on the left side subassembly **308**. These power and/or control interfaces on the left side subassembly **308** are connected to the drive shaft **316** and/or the fixed shaft **318** to provide power, control signaling, and/or other data to the laser head enclosure **406** (and the components contained therein).

(67) In some embodiments, power is provided to the laser head enclosure **406** (and the components contained therein) via the drive shaft **316** and/or the fixed shaft **318**, where (i) one of the drive shaft **316** or fixed shaft **318** is configured as an anode, and (ii) the other of the drive shaft **316** or fixed shaft **318** is configured as a cathode. For example, 5 volt DC power (or 12 volt DC power or any other suitable DC or AC voltage) is applied to one of the two shafts, and the other of the two shafts functions as a ground or return.

(68) In some embodiments, an interface on the left side subassembly **308** to a power source (e.g., a

power source located in control compartment **116**) is connected to a first graphite brush (not shown) on the left side subassembly **308** via a first power jumper cable. The first graphite brush on the left side subassembly **308** is electrically coupled to the fixed shaft **318**. A second graphite brush (not shown) on the carriage **302** (or alternatively on the laser head enclosure **406**) is also electrically coupled to the fixed shaft **318**. This second graphite brush on the carriage **302** (or laser head enclosure **406**) is connected to a laser head control board via a second power jumper cable. Thus, in operation, the power is routed from the power interface on the left side subassembly **308** to the laser head control board in the laser head enclosure **406** via a connection comprising the first power jumper cable, the first graphite brush, the fixed shaft **318**, the second graphite brush, and the second power jumper cable.

(69) In such embodiments where the power is provided to the laser head enclosure **406** via the fixed shaft **318** in manner described above, the ground or return is provided to the laser head enclosure **406** via the drive shaft **316**. A ground interface on the left side subassembly **308** to ground (e.g., a ground located in control compartment **116**) is connected to a third graphite brush (not shown) on the left side subassembly **308** via a first ground jumper cable. The third graphite brush on the left side subassembly **308** is electrically coupled to the drive shaft **316**. A fourth graphite brush (not shown) on the carriage **302** (or alternatively on the laser head enclosure **406**) is also electrically coupled to the drive shaft **316**. This fourth graphite brush on the carriage **302** (or laser head enclosure **406**) is connected to the laser head control board via a second ground jumper cable. Thus, in operation, the ground or return is routed from the laser head control board in the laser head enclosure **406** to the ground interface on the left side subassembly **308** via a connection comprising the first ground jumper cable, the third graphite brush, the drive shaft **316**, the fourth graphite brush, and the second ground jumper cable. Alternatively, the drive shaft **316** could be used for power and the fixed shaft **318** could be used for the ground or return.

(70) In some embodiments, control signaling to control the operation of components within the laser head enclosure **406** is also provided via the drive shaft **316** and fixed shaft **318**. For example, control signaling to/from components within the control compartment **116** can be transmitted to and/or received from components within the laser head enclosure **406** via carrier modulated signals transmitted along the fixed shaft **318** and/or drive shaft **316**. Additionally or alternatively, the components for controlling the operation of the gantry assembly **300/301** (e.g., one or more processors and other control components) could be located on the left side subassembly **308**. In some embodiments, control signaling can be transmitted to and/or received from components within the laser head enclosure **406** via carrier modulated signals transmitted along the same combination of jumpers and graphite brushes used for providing power and ground connections to the laser head enclosure **406** via the fixed shaft **318** and/or drive shaft **316**.

(71) Some embodiments that provide power, control signaling, and/or other data to components within the laser head enclosure **406** may additionally include (i) one or more high pass filters for use with coupling the control signal to the fixed shaft **318** and/or drive shaft **316**, and/or (ii) one or more low pass filters for use with coupling the power signal to the fixed shaft **318** and/or drive shaft **316**.

(72) Although the embodiments described above employ graphite brushes for coupling power and signaling to the fixed shaft **318** and drive shaft **316**, other suitable couplings could be used in addition to or instead of graphite brushes, including any type of electrical coupling now known or later developed that is suitable for transferring power and control signaling to/from the fixed and drive shafts.

(73) In still further embodiments, control signaling to control the operation of components within the laser head enclosure **406** could alternatively be provided via wireless transmission. For example, control signaling to/from components within the control compartment **116** can be wirelessly transmitted to and/or received from components within the laser head enclosure **406** via any wireless transmission protocol suitable for transmitting and receiving control signals for

controlling the operation of components within the laser head enclosure **406**.

(74) In still further embodiments, the laser head enclosure **406** may include one or more processors and tangible, non-transitory computer readable media with program instructions that, when executed by the one or more processors, configure the one or more processors to control the components of the gantry assembly **300/301**. Such embodiments may employ any of the arrangements of brushes, jumpers, and filters (in any suitable combination) described above for providing power, control signaling, and/or other data to/from the laser head enclosure **406**. In such embodiments, the one or more processors within the laser head enclosure **406** are configured to transmit control signals (e.g., wirelessly, via a flex/ribbon cable, and/or via structural components of the gantry assembly such as the drive shaft **316** and fixed shaft **318**) to the x-motor **312** and y-motor **314** on the left side subassembly **308**. In such embodiments, housing the one or more processors for operating and controlling the gantry assembly **300/301** and the components within the laser head enclosure **406** within the laser head enclosure **406** facilitates easy replacement and upgrade of the laser and other control components contained within the laser head enclosure **406**.

(75) In embodiments consistent with FIGS. **4A** and **4B**, the laser head enclosure **406** is detachable from the carriage **302** to facilitate maintenance, configuration, repair, replacement, and/or upgrading the laser head enclosure **406** and/or the laser head contained therein. However, in other embodiments, the laser head enclosure **406** may not be detachable from the carriage **302**.

(76) To facilitate correct alignment of the laser head enclosure **406** with the carriage **302** for attachment, detachment, re-attachment, etc. of the laser head enclosure **406** to the carriage **302**, the carriage **302** includes (i) a set of guide pins **402a**, **402b**, and **402c**, and (ii) a set of guide holes **403a**, **403b**, **403c**, and **403d**. In some embodiments, one or more of the guide pins **402a**, **402b**, and **402c** comprise magnetic guide pins, and/or one or more of the guide holes **403a**, **403b**, **403c**, and **403d** comprise magnetic guide holes. In operation, embodiments with magnetic guide pins and/or magnetic guide holes help to guide the laser head enclosure **406** into place during attachment and hold the laser head enclosure **406** securely in place during operation.

(77) In the example embodiment shown in FIG. **4A**, the guide holes **403a**, **403b**, **403c**, and **403d** in the carriage **302** are arranged to receive corresponding laser head enclosure guide pins (not shown) extending from the laser head enclosure **406**. Likewise, laser head guide holes (not shown) are arranged to receive the guide pins **402a**, **402b**, and **402c** extending from the carriage **302**. This guide pin and guide hole arrangement ensures that the laser head enclosure **406** is reliably attached, affixed, mated, or otherwise joined to the carriage **302**, and also ensures that the laser head enclosure **406** is held securely in place during operation of the laser.

(78) In some embodiments consistent with the examples shown in FIGS. **4A**, **4B**, and **4D**, the carriage **302** includes two openings: (i) a fixed shaft opening **350** and (ii) a drive shaft opening **352**. The fixed shaft **318** extends through the fixed shaft opening **350** of the carriage **302**, and the drive shaft **316** extends through the drive shaft opening **352** of the carriage **302**. The drive shaft opening **352** of the carriage **302** is configured to allow the drive shaft **316** to axially rotate within drive shaft opening **352**. In some embodiments, the circumference of the drive shaft opening **352** is sufficiently larger than the circumference of the drive shaft **316**. For example, in some embodiments, the circumference of the drive shaft opening **352** is between 50-150 microns larger than the circumference of the drive shaft **316**.

(79) In operation, the fixed shaft **318** and the drive shaft **316** are arranged to guide movement of the carriage **302** in the x-direction by allowing the carriage **302** to slide back-and-forth along the fixed shaft **318** and the drive shaft **316**. In this manner, the fixed shaft **318** and drive shaft **316** function like guide rails for the carriage **302** to facilitate accurate positioning of the carriage **302** (and the laser head enclosure **304** affixed thereto). In some embodiments, the carriage **302** includes linear bearings within the fixed shaft opening **350** and/or drive shaft opening **352** to facilitate smooth sliding of the carriage **302** along the fixed shaft **318** and the drive shaft **316**.

(80) FIGS. **4A**, **4B**, and **4D** show the drive shaft **316** and the fixed shaft **318** as substantially the

same circumference. But in other embodiments, the drive shaft **316** and the fixed shaft **318** may be different sizes. In some embodiments, the fixed shaft **318** may be larger than the drive shaft **316**, e.g., 25%-150% larger than the drive shaft **316**. Alternatively, the drive shaft **316** may be larger than the fixed shaft **318**.

(81) Alternatively, some embodiments may implement a drive mechanism different than a drive shaft, such as a drive chain, drive belt, or any other alternative drive mechanism now known or later developed that is suitable for transferring mechanical energy from one or more gears on the left side subassembly **308** (FIGS. 3A, 3B) to one or more gears on the right side subassembly **306** (FIGS. 3A, 3B). In some embodiments, at least a portion of the alternative drive mechanism may be contained within a housing over which the carriage **302** is configured to slide. In some embodiments, at least a portion of the drive shaft **316** may be contained within a housing over which the carriage **302** is configured to slide.

(82) FIG. 4C shows an angle adapter **408** for use with the carriage **302**. The angle adapter **408** is configured to attach to the top of the carriage **302**. As shown in FIG. 4D, when the laser head enclosure **406** is attached to the angle adapter **408**, the laser head within the laser head enclosure **406** is positioned to fire in an outward direction rather than a downward direction between the drive shaft **316** and the fixed shaft **318** as shown in FIGS. 4A and 4B.

(83) To facilitate attachment of the laser head enclosure **406** to the angle adapter **408**, the angle adapter **408** includes (i) guide pins **410a**, **410b**, and **410c**, and (ii) guide holes **411a**, **411b**, **411c**, and **411d**. In some embodiments, one or more of the guide pins **410a**, **410b**, and **410c** comprise magnetic guide pins, and/or one or more of the guide holes **411a**, **411b**, **411c**, and **411d** comprise magnetic guide holes.

(84) In operation, the guide pins **410a**, **410b**, and **410c**, and the guide holes **411a**, **411b**, **411c**, and **411d** of the angle adapter **408** perform substantially the same function as the guide pins **402a**, **402b**, and **402c**, and the guide holes **403a**, **403b**, **403c**, and **403d** of the carriage **302**. In particular, the guide holes **411a**, **411b**, **411c**, and **411d** in the angle adapter **408** are arranged to receive corresponding laser head enclosure guide pins (not shown) extending from the laser head enclosure **406**. Likewise, laser head guide holes (not shown) are arranged to receive the guide pins **410a**, **410b**, and **410c** extending from the angle adapter **408**. This guide pin and guide hole arrangement ensures that the laser head enclosure **406** is reliably attached, affixed, mated, or otherwise joined to the angle adapter **408**.

(85) In some embodiments, when the laser head enclosure **406** is attached to the carriage **302** via the angle adapter **408** as shown in FIGS. 4C and 4D, the laser within the laser head enclosure **406** is positioned to fire in a direction that is substantially perpendicular to the direction that the laser fires when the laser head enclosure **406** is attached to the carriage **302** directly as shown in the FIGS. 4A and 4B. In some embodiments, instead of 90° relative to the downward laser firing direction shown in FIGS. 4A and 4B, the angle adapter **408** shown in FIGS. 4C and 4D may position the laser to fire at a different angle that is greater or less than 90° relative to the downward laser firing direction shown in FIGS. 4A and 4B. In some embodiments, the carriage **302** may be arranged to accommodate several different angle adaptors that each are configured to fire the laser at different angles relative to the downward laser firing direction shown in FIGS. 4A and 4B. In some embodiments, the angle adapter **408** is adjustable to accommodate different firing angles.

(86) In operation, the angle adapter **408** enables the CNC machine to accommodate material that might be taller than the space between the floor (sometimes referred to as the material bed) of the interior area **104** (FIG. 1) and the joining subassembly **310** (FIGS. 3A and 3B) of the gantry assembly **300/301**. In some embodiments, the material to be processed by the CNC machine sits on a rotary adapter (not shown) which can turn (i.e., spin) the material so that a sideways-firing laser (e.g., as shown in FIGS. 4C and 4D) can apply laser energy to the material in both x- and y-dimensions.

(87) In some embodiments of this sideways-firing configuration, the gantry assembly **300/301**

(FIGS. 3A and 3B) additionally includes a soft stop feature configured to stop movement of the gantry assembly components in the y dimension so that components of the gantry assembly (e.g., the carriage **302**, joining subassembly **310**, the right side subassembly **306**, and/or the left side subassembly **308**) do not run into the material being processed by the CNC machine. In some embodiments, this soft stop feature is implemented via one or more cameras and/or sensors on the gantry assembly and/or disposed within the interior space **104** of the housing **102**. In operation, the one or more cameras and/or other sensors are configured to detect or otherwise sense the position of the material being machined by the CNC machine. Image and/or sensor data from these one or more cameras and/or sensors is provided to one or more processors, which in turn use the image and/or sensor data to control the movement and positioning of the gantry assembly.

(88) In some alternative embodiments, the joining assembly **310** may comprise a single housing that (i) rigidly connects the left side subassembly **308** with the right side subassembly **306** in a manner similar to the fixed shaft **318**, and (ii) houses a drive mechanism, such drive shaft **316** or an alternative drive mechanism as described above. In other alternative embodiments, instead of (or in addition to) the fixed shaft **318**, the joining assembly **310** may include a rigid beam, a rigid plate, or other rigid structure to connect the right side subassembly **306** with the left side subassembly **308**. In still further alternative embodiments, the joining subassembly **310** may include multiple fixed shafts (or similar rigid structures) rather than a single fixed shaft.

(89) The right side subassembly **306** and the left side subassembly **308** are shown and described with reference to FIGS. 3A and 3B as separate subassemblies connected by the joining assembly **310**. But in some embodiments, the right side subassembly **306**, left side subassembly **308**, and joining assembly **310** (or at least the fixed shaft **318** component thereof) may be integrated into a single component made from metal, plastic, or other suitably rigid material. However, in some embodiments, it may be advantageous for the right side subassembly **306**, left side subassembly **308**, and joining assembly **310** (or at least the fixed shaft **318** component thereof) to be manufactured as separate components and then connected together to form a structure sufficiently rigid to support positioning of the carriage **302** within the x-y plane as described above.

(90) As shown in FIGS. 3A and 3B and described herein, the x-motor **312** and y-motor **314** that operate gantry assembly **300/301** are both located on the left side subassembly **308**. In some embodiments, the x-motor **312** and the y-motor **314** may be alternatively mounted on the right side subassembly **306**. Either way, because the x-motor **312** and y-motor **314** are located together on the same subassembly (either the right or left), the position of the x-motor **312** relative to the position of the y-motor **314** remains constant during (i) movement of the carriage **302** along the length of the joining assembly **310** in the x-direction between the right side subassembly **306** and the left side subassembly **308** and (ii) movement of the right side subassembly **306** and the left side subassembly **308** together in the y-direction.

(91) Locating the x-motor **312** and the y-motor **314** on the left side subassembly **308** (or alternatively on the right side subassembly **306**) provides technical and operational advantages over typical gantry assemblies. For example, locating both the x-motor **312** and the y-motor **314** on the same subassembly enables delivery of power and control signaling to the gantry assembly **300/301** via a single wiring harness. Such a configuration reduces the amount of wiring required to power and control the motors that operate the gantry assembly **300/301** and enables the gantry assembly **300/301** to be powered and controlled from only one side of the gantry assembly **300/301**.

Additionally, locating the x-motor **312** on the left side subassembly **308** (or alternatively on the right side subassembly **306**) with the y-motor **314** enables the gantry assembly **300/301** to employ a lightweight carriage **302** as compared to alternative implementations that may locate an x-motor on the carriage **302** to control movement of the carriage **302** (and the laser head enclosure **406**)

(92) The configuration of having both the x-motor **312** and the y-motor **314** located together on the same side subassembly also eliminates the need for a shaft **236** (FIG. 2B) to connect rail-mounted y-direction drive components that would otherwise obstruct material pass-through areas (e.g., pass-

through slots **120** and **121** or similar pass through areas) or other regions within the operating area of the CNC machine, and limit the size of the materials that the CNC machine can process. Thus, CNC machines that employ gantry assembly **300** (or gantry assembly **301**) can accommodate larger materials to process as compared to typical gantry assemblies that incorporate a shaft **236** (FIG. 2B) or similar drive shaft and/or stabilization shaft arrangements.

(93) Further, the configuration of having all the motors on a single side subassembly of the gantry assembly **300/301** also simplifies the initial assembling and later repairing of the CNC machine as compared to existing configurations. For example, the housing **102** can be manufactured separately from the gantry assembly **300/301**, where all of the active mechanical components are on the gantry assembly **300/301**. This enables the gantry assembly **300/301** to be mated to the housing **102** in the final manufacturing steps without requiring any electromechanical interfaces between the gantry assembly **300/301** and the housing **102**. For example, mating the gantry assembly **300/301** with the housing **102** could include simply bolting (or otherwise attaching) the y-rails **326** and **328** to the housing **102** and connecting a single wiring harness to the left side subassembly **308** to power and control the x-motor **312** and y-motor **314** located on the left side subassembly **308**.

(94) Additionally, the gantry assembly **300/301** can later be removed in one piece for repair or servicing, and then re-installed in the housing **102** without having to disconnect/reconnect electromechanical interfaces. For example, removing the gantry assembly **300/301** from the housing **102** for repair or servicing could include unplugging a single wiring harness and unbolting (or otherwise detaching) the y-rails **326** and **328** from the housing **102**. And replacing the gantry assembly **300/301** in the housing **102** after servicing and/or repair could include re-bolting (or otherwise re-attaching) the y-rails **326** and **328** to the enclosure and reconnecting the wiring harness.

(95) In some embodiments, the gantry structure (i.e., the combination of the right side subassembly **306**, the left side subassembly **308**, and the joining subassembly **310** that move together as a single component or structure) of the gantry assembly **300/301** is removable from the y-rails **326** and **328** to facilitate servicing or repair. In such embodiments, the gantry structure can be removed from the y-rails **326** and **328** in one piece for servicing or repair, and then re-attached onto the y-rails **326** and **328** without having to disconnect/reconnect the y-rails **326** and **328** to/from the enclosure. For example, removing the gantry structure (i.e., the combination of the right side subassembly **306**, the left side subassembly **308**, and the joining subassembly **310** that move together as a single component or structure) from the y-rails **326** and **328** for repair or servicing could include unplugging a single wiring harness and detaching the gantry structure from the y-rails **326** and **328** while the y-rails **326** and **328** remain attached the housing **102**. And replacing the gantry structure after servicing and/or repair could include re-attaching the gantry structure to the y-rails **326** and **328** and re-connecting the wiring harness.

(96) In some embodiments, rather than using a ribbon cable or other type of wiring harness to provide power and control signaling to the laser head, structural elements of the gantry (such as, for example, the drive shaft **316** and fixed shaft **318**) may be used to provide power and/or transmit control signals.

(97) Further, the design of the gantry assembly **300/301** facilitates easy upgrading or changing of the gantry assembly in the future. For example, removing the gantry assembly **300/301** from the housing **102** for upgrade or swap-out could include unplugging a single wiring harness and unbolting (or otherwise un-attaching) the y-rails **326** and **328** from the housing **102**. And installing an upgrades or swap-out gantry assembly in the housing **102** could include bolting (or otherwise attaching) the y-rails of the upgraded or swapped out gantry assembly to the enclosure and connecting the wiring harness thereto. For example, when a newer gantry assembly with newer lasers, newer laser lenses, more precise range of travel, better gearing, or other improvements is available, the older gantry assembly can be removed and replaced with the upgraded gantry assembly. Similarly, when a different gantry assembly with different tools (e.g., a different laser, a

mechanical cutter, or other different machining tool) is desired for a particular project, the existing gantry assembly can be swapped out with the different gantry assembly. And when the projecting requiring the different gantry assembly is complete, the original gantry assembly can be swapped back in to replace the different gantry assembly.

V. Conclusions

(98) The description above discloses, among other things, various example systems, configurations, apparatus, and articles of manufacture. The examples disclosed and described herein are merely illustrative and should not be considered as limiting. Additionally, references herein to “embodiment” generally means that a particular feature, structure, or characteristic described in connection with the embodiment can be included in at least one example embodiment of an invention. The appearances of this phrase in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments mutually exclusive of other embodiments. As such, the embodiments described herein, explicitly and implicitly understood by one skilled in the art, can be combined with other embodiments.

(99) Numerous specific details are set forth to provide a thorough understanding of the present disclosure. However, it is understood to those skilled in the art that certain embodiments of the present disclosure can be practiced without certain, specific details. In other instances, well known methods, procedures, components, and circuitry have not been described in detail to avoid unnecessarily obscuring aspects of the embodiments. Accordingly, the scope of the present disclosure is defined by the appended claims rather than the foregoing description of embodiments.

(100) When any of the appended claims are read to cover a purely software and/or firmware implementation, at least one of the elements in at least one example is hereby expressly defined to include a tangible, non-transitory medium such as a memory, including Read Only Memory (ROM), Random Access Memory (RAM), or other type of memory now known or later developed that is sufficient for storing software and/or firmware containing program instructions for execution by one or more processors.

Claims

1. A system comprising a gantry assembly, wherein the gantry assembly comprises: a carriage configured to accommodate a laser head; a first movable subassembly; and a second movable subassembly coupled to the first movable subassembly via a joining subassembly extending between the first movable subassembly and the second movable subassembly, wherein the second movable subassembly comprises (i) a first motor configured to control a position of the carriage along the joining subassembly along a first axis between the first movable subassembly and the second movable subassembly and (ii) a second motor configured to control a position of the first movable subassembly and the second movable subassembly together along a second axis substantially perpendicular to the first axis, thereby controlling a position of the carriage along the second axis; and wherein the joining subassembly comprises a drive mechanism operated by the second motor, wherein the drive mechanism is arranged to facilitate movement of the first movable subassembly together with the second movable subassembly along the second axis, wherein the drive mechanism operated by the second motor comprises a drive shaft that extends along at least a portion of the joining subassembly and is arranged to facilitate movement of the first movable subassembly together with the second movable subassembly along the second axis, and wherein the first movable subassembly further comprises a first gear connected to the drive shaft, wherein the second movable subassembly further comprises a second gear connected to the drive shaft and coupled to the second motor via a linkage, and wherein the second motor is configured to (i) control movement of the second gear via the linkage and (ii) control movement of the first gear by axially rotating the second gear about the first axis, thereby axially rotating the drive shaft and the first gear connected thereto about the first axis.

2. The system of claim 1, wherein a position of the first motor relative to a position of the second motor remains constant during (i) movement of the carriage along the first axis between the first movable subassembly and the second movable subassembly and (ii) movement of the first movable subassembly and the second movable subassembly together along the second axis.
3. The system of claim 1, wherein the joining subassembly further comprises a fixed shaft connecting the first movable subassembly to the second movable subassembly.
4. The system of claim 1, further comprising: a first anti-lift stabilization mechanism located on the first movable subassembly and configured to move with the first movable subassembly; and a second anti-lift stabilization mechanism located on the second movable subassembly and configured to move with the second movable subassembly; and wherein the first anti-lift stabilization mechanism and the second anti-lift stabilization mechanism are configured to constrain movement of the first movable subassembly and the second movable subassembly in a third axis during operation, and wherein the third axis is substantially perpendicular to both the first axis and the second axis.
5. The system of claim 1, wherein the joining subassembly comprises a fixed shaft connecting the first movable subassembly to the second movable subassembly, and wherein the fixed shaft is substantially parallel to the drive shaft.
6. The system of claim 5, wherein the fixed shaft extends through a first opening of the carriage, wherein the drive shaft extends through a second opening of the carriage, and wherein the fixed shaft and the drive shaft are arranged to guide movement of the carriage along the first axis.
7. The system of claim 6, wherein the second opening of the carriage is configured to allow the drive shaft to axially rotate about the first axis within the second opening.
8. The system of claim 1, wherein the first movable subassembly comprises sufficient weight arranged to counteract a lifting force imposed upon the first movable subassembly via a rotation of the drive shaft.
9. The system of claim 1, wherein the second movable subassembly further comprises a third gear coupled to the first motor via a second linkage, and wherein the first motor is configured to control the position of the carriage in the first axis along the joining subassembly via the second linkage and the third gear.
10. The system of claim 1, wherein the first motor comprises a first stepper motor, and wherein the second motor comprises a second stepper motor.
11. The system of claim 1, further comprising: a housing comprising at least four interior walls defining an interior space; a first rail, wherein the first rail is proximate to and extends substantially along a first interior wall of the housing, and wherein the first movable subassembly is configured to move along the second axis via the first rail; and a second rail, wherein the second rail is proximate to and extends substantially along a second interior wall of the housing opposite the first interior wall of the housing, wherein the second movable subassembly is configured to move along the second axis via the second rail, wherein the second rail is substantially parallel to the first rail.
12. The system of claim 11, further comprising: a first interface between the first movable subassembly and the first rail, wherein the first interface is configured to provide a first preload force between the first movable subassembly and the first rail; and a second interface between the second movable subassembly and the second rail, wherein the second interface is configured to provide a second preload force between the second movable subassembly and the second rail.
13. The system of claim 11, wherein the gantry assembly further comprises: a first belt-and-pulley arrangement configured to facilitate movement of the first movable subassembly along the second axis via the first rail; and a second belt-and-pulley arrangement configured to facilitate movement of the second movable subassembly along the second axis via the second rail parallel to the first rail.
14. The system of claim 11, wherein the gantry assembly further comprises: a first rack gear assembly configured to facilitate movement of the first movable subassembly along the second axis

via the first rail; and a second rack gear assembly configured to facilitate movement of the second movable subassembly along the second axis via the second rail parallel to the first rail.

15. The system of claim 11, wherein the housing further comprises a pass through area arranged to allow materials to be moved into and out of the interior space free from obstruction caused by the gantry assembly.

16. The system of claim 11, further comprising: a wiring harness arranged to provide power and control signals to the first motor and the second motor from outside of the housing.

17. The system of claim 11, further comprising: a wiring harness connected to the second movable subassembly and configured to provide power and control signals to both the first motor and the second motor; and wherein the gantry assembly is configured to be removable as a single structure by (i) disconnecting the wiring harness, and (ii) detaching the first movable subassembly and second movable subassembly from the first rail and the second rail, respectively, while the first rail and the second rail remain attached to the housing; and wherein the gantry assembly is configured to be re-installable as a single structure by (i) reattaching the first movable subassembly and second movable subassembly to the first rail and the second rail, respectively, and (ii) reconnecting the wiring harness.

18. The system of claim 1, further comprising one or more stabilization assemblies configured to constrain movement of the gantry assembly along a third axis that is substantially perpendicular to both the first axis and the second axis.

19. The system of claim 13, further comprising: a third belt-and-pulley arrangement configured to facilitate movement of the carriage along the first axis via the joining subassembly extending between the first movable subassembly and the second movable subassembly.

20. The system of claim 1, further comprising: one or more processors; a tangible, non-transitory computer-readable memory; and program instructions stored on the tangible, non-transitory computer-readable memory, wherein the program instructions are executable by the one or more processors to cause the system to: control a position of the laser head along the first axis by controlling the first motor; and control a position of the laser head along the second axis by controlling the second motor.
